Internal Migration Analysis Through Remittance Data in Honduras

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1 Data Description

The dataset `remesas_year_munic.csv` contains individual-level remittance records aggregated by origin (birth) and destination (receipt) municipality in Honduras, spanning 2020–2024. Each row represents a unique combination of:

Column	Description
`Year`	Year of remittance receipt
`codDeptoNac`	Department code where the person **was born**
`codMunNac`	Municipality code where the person **was born**
`codDepto`	Department code where the person **received** the remittance
`codMun`	Municipality code where the person **received** the remittance
`in_city`	**1** if the person received the remittance in the **same municipality** where they were born
`capital`	**1** if the person received the remittance in **Tegucigalpa (Distrito Central)** but was **not** born there
`sps`	**1** if the person received the remittance in **San Pedro Sula** but was **not** born there
`other_main`	**1** if the person received the remittance in a city **different from Tegucigalpa or San Pedro Sula** and was **not** born there
`N`	Number of remittance beneficiaries
`USD`	Total remittance amount in US dollars

1.1 Interpretation as a Migration Proxy

When a person **was born** in one municipality but **receives remittances** in a different municipality, this strongly suggests that the person has **migrated internally** from their birth municipality to the receipt municipality. The flags `capital`, `sps`, and `other_main` capture this migration signal:

- **`capital = 1`**: Internal migrants who moved **to Tegucigalpa** (department 8, municipality 1 — Francisco Morazán, Distrito Central).
- **`sps = 1`**: Internal migrants who moved **to San Pedro Sula** (department 5, municipality 1 — Cortés, San Pedro Sula).
- **`other_main = 1`**: Internal migrants who moved to **other cities** that are neither Tegucigalpa nor San Pedro Sula.

This allows us to study **rural-to-urban migration** patterns by analysing which departments and municipalities export people toward the two main economic hubs of Honduras.

---

2 Setup

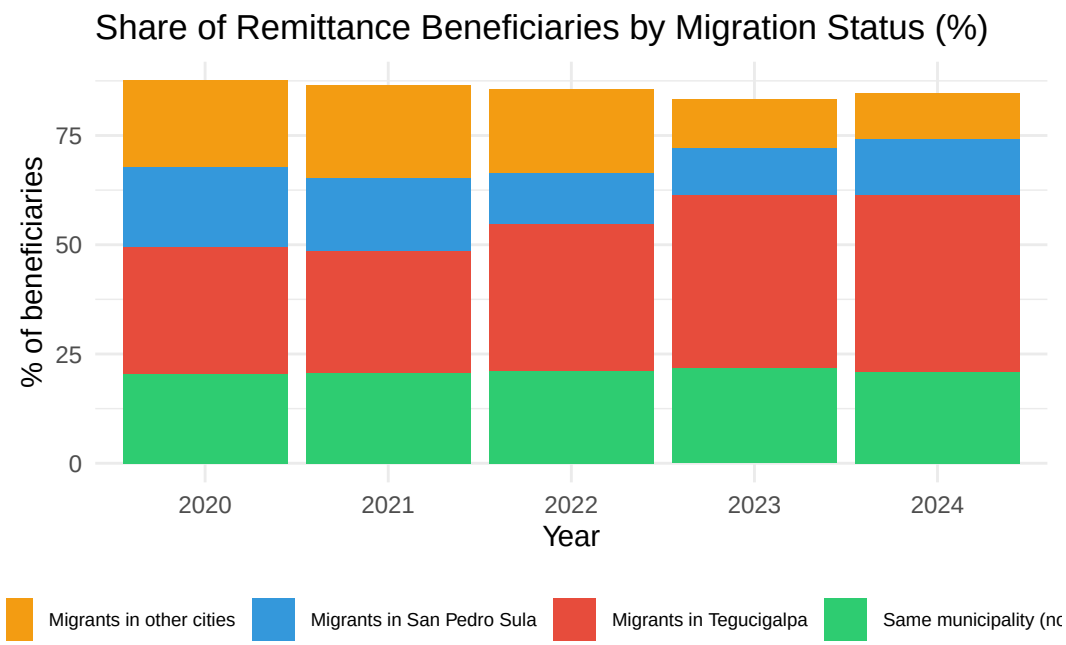
```r
library(tidyverse)
library(kableExtra)
library(scales)
library(patchwork)

df <- read_csv("remesas_year_munic.csv", show_col_types = FALSE)

# Department code-to-name mapping (from OCHA admin boundaries)
dept_names <- tibble(
  code = 1:18,
  dept_name = c(
    "Atlántida", "Colón", "Comayagua", "Copán",
    "Cortés", "Choluteca", "El Paraíso",
    "Francisco Morazán", "Gracias a Dios", "Intibucá",
    "Islas de la Bahía", "La Paz", "Lempira",
    "Ocotepeque", "Olancho", "Santa Bárbara",
    "Valle", "Yoro"
  )
)

df <- df |>
  left_join(dept_names, by = c("codDeptoNac" = "code")) |>
  rename(dept_born = dept_name) |>
  left_join(dept_names, by = c("codDepto" = "code")) |>
  rename(dept_recv = dept_name)
```

3 1 — Aggregate Migration Flows to Tegucigalpa and San Pedro Sula

How many remittance beneficiaries in Tegucigalpa and San Pedro Sula were **not born there**? This section quantifies the yearly volume of "migrant" beneficiaries flowing into the two main cities versus those who stayed in their birth municipality.

```r
yearly <- df |>
  group_by(Year) |>
  summarise(
    total_N     = sum(N),
    total_USD   = sum(USD),
    in_city_N   = sum(N[in_city == 1]),
    capital_N   = sum(N[capital == 1]),
    sps_N       = sum(N[sps == 1]),
    other_N     = sum(N[other_main == 1]),
    capital_USD = sum(USD[capital == 1]),
    sps_USD     = sum(USD[sps == 1]),
```

```r
    .groups = "drop"
  ) |>
  mutate(
    capital_pct = round(capital_N / total_N * 100, 1),
    sps_pct     = round(sps_N / total_N * 100, 1),
    other_pct   = round(other_N / total_N * 100, 1),
    in_city_pct = round(in_city_N / total_N * 100, 1)
  )

yearly_long <- yearly |>
  select(Year, in_city_pct, capital_pct, sps_pct, other_pct) |>
  pivot_longer(-Year, names_to = "category", values_to = "pct") |>
  mutate(category = factor(category,
    levels = c("other_pct", "sps_pct", "capital_pct", "in_city_pct"),
    labels = c("Migrants in other cities", "Migrants in San Pedro Sula",
               "Migrants in Tegucigalpa", "Same municipality (no migration)")))

ggplot(yearly_long, aes(x = factor(Year), y = pct, fill = category)) +
  geom_col() +
  scale_fill_manual(values = c(
    "Same municipality (no migration)" = "#2ecc71",
    "Migrants in Tegucigalpa" = "#e74c3c",
    "Migrants in San Pedro Sula" = "#3498db",
    "Migrants in other cities" = "#f39c12")) +
  labs(title = "Share of Remittance Beneficiaries by Migration Status (%)",
       x = "Year", y = "% of beneficiaries", fill = NULL) +
  theme_minimal() +
  theme(legend.position = "bottom", legend.text = element_text(size = 7))
```



The stacked bar chart above decomposes the universe of remittance beneficiaries into four mutually exclusive groups each year. The dominant green segment ("Same municipality") represents people who still live where they were born — they have not migrated. The red (Tegucigalpa) and blue (San Pedro Sula) segments capture internal migrants who moved to the two main cities, while the orange segment captures migrants to other urban centres. The relative stability of these shares over 2020–2024 indicates that the overall migration structure has not shifted drastically, although small year-to-year changes may reflect economic shocks (e.g., the post-COVID recovery in 2021–2022) or data coverage adjustments.

```
# Absolute numbers table
yearly |>
  select(Year, total_N, in_city_N, capital_N, sps_N, other_N,
         capital_pct, sps_pct) |>
  rename(
    Total = total_N, `Same city` = in_city_N,
    `Tegucigalpa (migrants)` = capital_N,
    `SPS (migrants)` = sps_N,
    `Other (migrants)` = other_N,
    `Tegu %` = capital_pct, `SPS %` = sps_pct
  ) |>
  kbl(booktabs = TRUE, format.args = list(big.mark = ",")) |>
  kable_styling(latex_options = c("striped", "hold_position"), font_size = 9)
```

The table above provides the raw numbers behind the stacked bar chart. Key columns to note: **Total** is the count of all remittance beneficiaries nationwide; **Tegucigalpa (migrants)** and **SPS (migrants)** are the subset who were born elsewhere but receive remittances in each city; **Tegu %** and **SPS %** express those migrants as a share of the national total. Together, migrants to the two

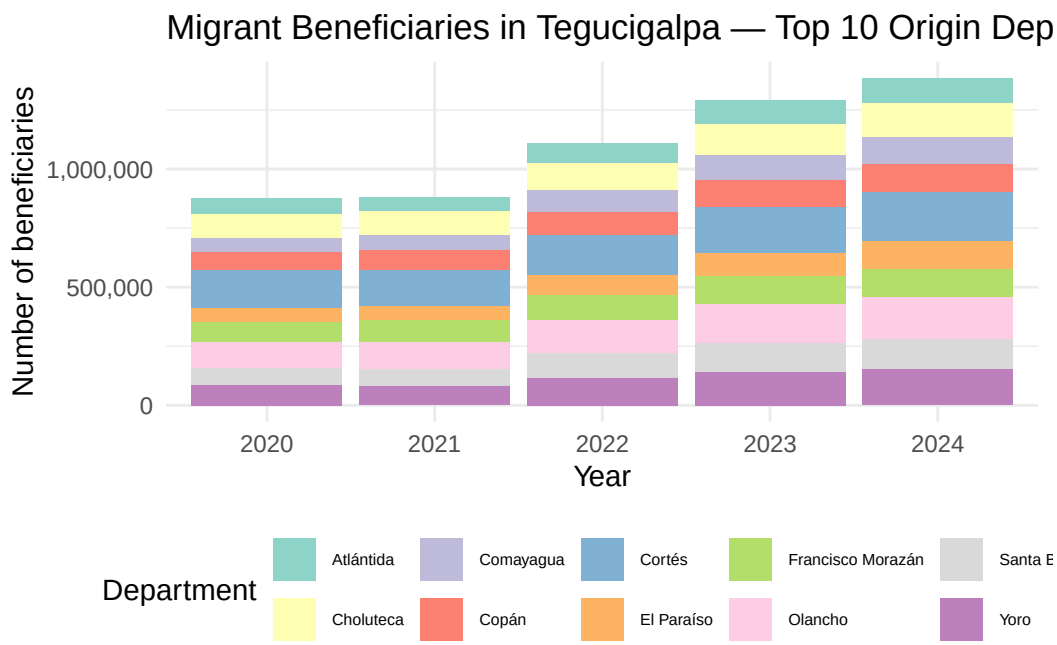
Year	Total	Same city	Tegucigalpa (migrants)	SPS (migrants)	Other (migrants)	Tegu %	SPS %
2,020	3,840,646	792,565	1,110,753	701,537	756,649	28.9	18.3
2,021	4,044,259	838,601	1,124,382	681,925	858,456	27.8	16.9
2,022	4,262,915	898,563	1,433,904	502,274	808,678	33.6	11.8
2,023	4,234,460	927,843	1,675,233	453,049	475,611	39.6	10.7
2,024	4,419,824	924,911	1,792,728	565,305	457,874	40.6	12.8

main cities typically account for a notable fraction of all beneficiaries, underscoring the importance of rural-to-urban migration in Honduras's remittance landscape.

4 2 — Origin Departments of Migrants in Tegucigalpa

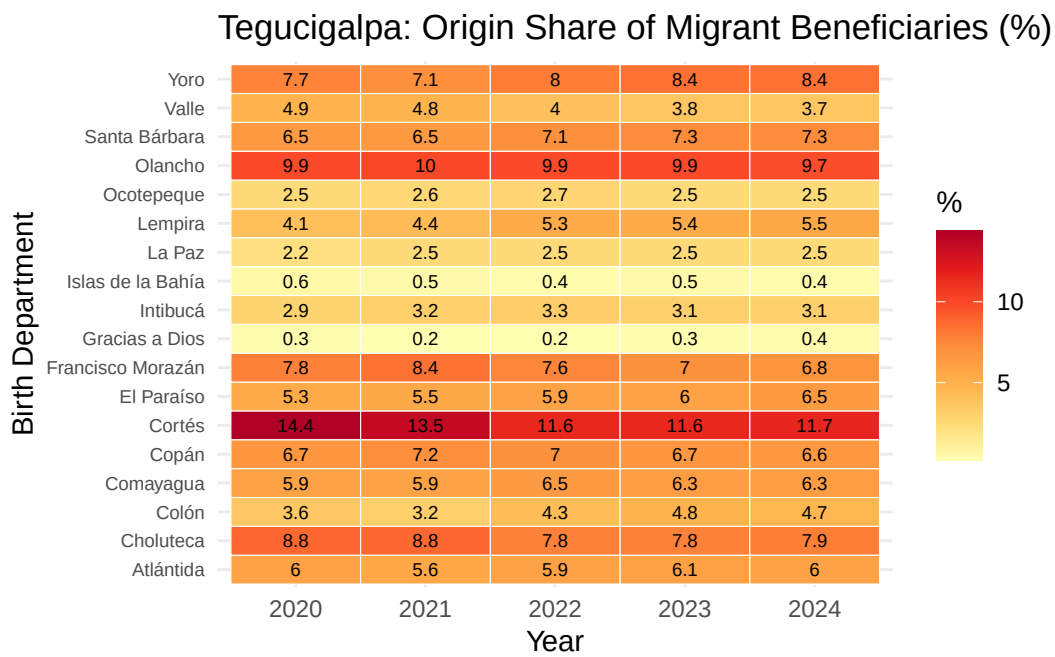
Which departments **export** the most people to the capital city?

```
cap_migrants <- df |> filter(capital == 1)

cap_by_dept <- cap_migrants |>
  group_by(Year, codDeptoNac, dept_born) |>
  summarise(N = sum(N), USD = sum(USD), .groups = "drop")

# Top 10 departments overall
top_depts_cap <- cap_by_dept |>
  group_by(dept_born) |>
  summarise(total = sum(N), .groups = "drop") |>
  slice_max(total, n = 10) |>
  pull(dept_born)

cap_top <- cap_by_dept |> filter(dept_born %in% top_depts_cap)

ggplot(cap_top, aes(x = factor(Year), y = N, fill = dept_born)) +
  geom_col() +
  scale_fill_brewer(palette = "Set3") +
  scale_y_continuous(labels = comma) +
  labs(title = "Migrant Beneficiaries in Tegucigalpa \u2014 Top 10 Origin Departments",
       x = "Year", y = "Number of beneficiaries", fill = "Department") +
  theme_minimal() +
  theme(legend.position = "bottom", legend.text = element_text(size = 6))
```



This stacked bar chart shows the top 10 origin departments for migrants living in Tegucigalpa. Each colour band represents one department, and the total height of the bar gives the aggregate volume of migrant beneficiaries for that year. Departments that consistently occupy wide bands — such as Olancho, El Paraíso, and Choluteca — are chronic "exporters" of people to the capital. Narrower bands for departments like Islas de la Bahía or Gracias a Dios reflect their smaller population bases and greater distance from Tegucigalpa.

```
# Heatmap: share of each origin department across years
cap_heat <- cap_by_dept |>
  group_by(dept_born, Year) |>
  summarise(N = sum(N), .groups = "drop") |>
  group_by(Year) |>
  mutate(pct = round(N / sum(N) * 100, 1)) |>
  ungroup()

ggplot(cap_heat, aes(x = factor(Year), y = dept_born, fill = pct)) +
  geom_tile(color = "white") +
  geom_text(aes(label = pct), size = 2.5) +
  scale_fill_distiller(palette = "YlOrRd", direction = 1) +
  labs(title = "Tegucigalpa: Origin Share of Migrant Beneficiaries (%)",
       x = "Year", y = "Birth Department", fill = "%") +
  theme_minimal() +
  theme(axis.text.y = element_text(size = 7))
```

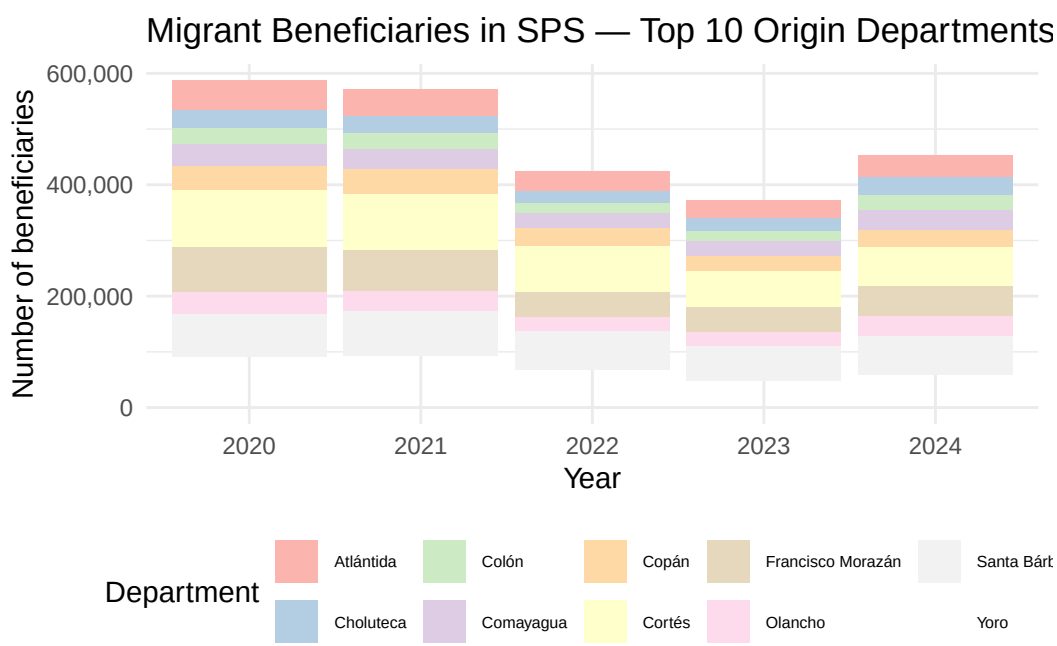


The heatmap shows each department's **percentage share** of all migrant beneficiaries in Tegucigalpa for every year. Darker cells indicate departments that contribute a larger fraction of the capital's migrants. This view normalises for different yearly totals, making it easy to spot structural shifts: if a department's share rises over time, it is becoming a relatively more important source of migration to the capital; a declining share may indicate that outflows are being redirected to San Pedro Sula or other cities, or that local conditions have improved.

5 3 — Origin Departments of Migrants in San Pedro Sula

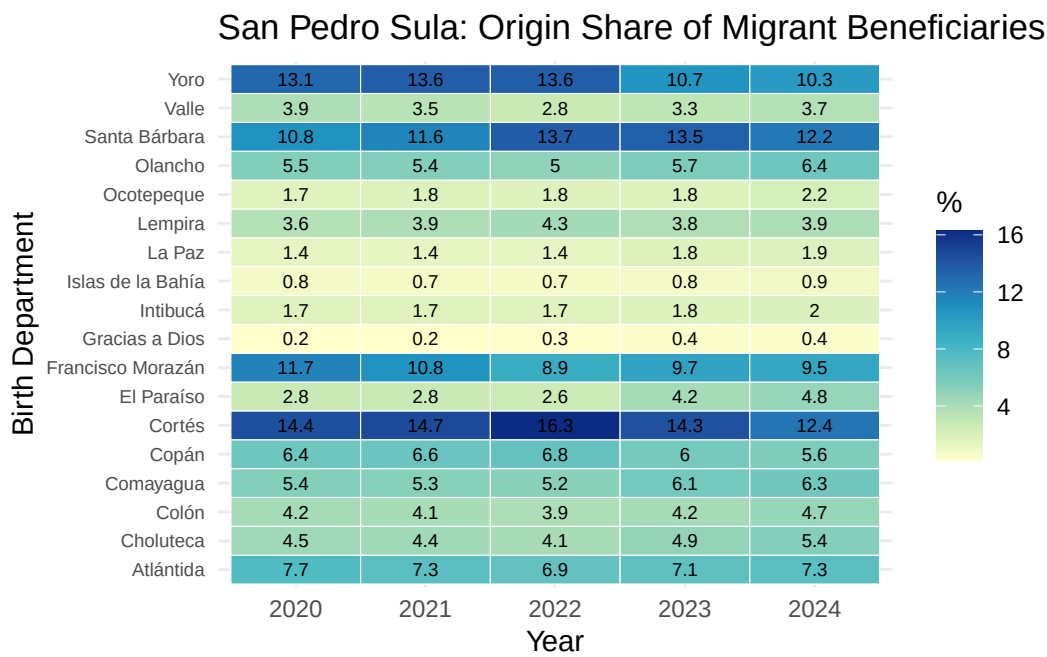
```
sps_migrants <- df |> filter(sps == 1)

sps_by_dept <- sps_migrants |>
  group_by(Year, codDeptoNac, dept_born) |>
  summarise(N = sum(N), USD = sum(USD), .groups = "drop")

top_depts_sps <- sps_by_dept |>
  group_by(dept_born) |>
  summarise(total = sum(N), .groups = "drop") |>
  slice_max(total, n = 10) |>
  pull(dept_born)

sps_top <- sps_by_dept |> filter(dept_born %in% top_depts_sps)

ggplot(sps_top, aes(x = factor(Year), y = N, fill = dept_born)) +
  geom_col() +
  scale_fill_brewer(palette = "Pastel1") +
```

```
  scale_y_continuous(labels = comma) +
  labs(title = "Migrant Beneficiaries in SPS \u2014 Top 10 Origin Departments",
       x = "Year", y = "Number of beneficiaries", fill = "Department") +
  theme_minimal() +
  theme(legend.position = "bottom", legend.text = element_text(size = 6))
```



The same stacked bar is now shown for San Pedro Sula. The origin department mix is noticeably different from Tegucigalpa: western departments such as Santa Bárbara, Copán, and Lempira tend to feature more prominently, reflecting SPS's role as the economic hub of the **Sula Valley** and its proximity to the western highlands. Cortés itself is excluded as the "home" department, so the chart focuses on true inter-department migrants.

```
sps_heat <- sps_by_dept |>
  group_by(dept_born, Year) |>
  summarise(N = sum(N), .groups = "drop") |>
  group_by(Year) |>
  mutate(pct = round(N / sum(N) * 100, 1)) |>
  ungroup()

ggplot(sps_heat, aes(x = factor(Year), y = dept_born, fill = pct)) +
  geom_tile(color = "white") +
  geom_text(aes(label = pct), size = 2.5) +
  scale_fill_distiller(palette = "YlGnBu", direction = 1) +
  labs(title = "San Pedro Sula: Origin Share of Migrant Beneficiaries (%)",
       x = "Year", y = "Birth Department", fill = "%") +
  theme_minimal() +
  theme(axis.text.y = element_text(size = 7))
```



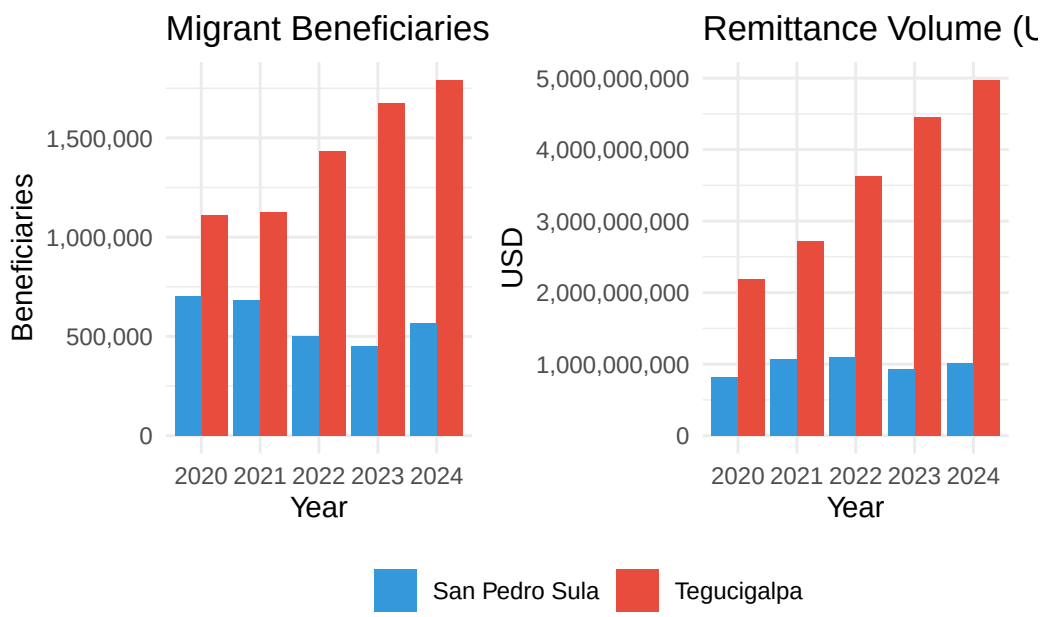
As with the Tegucigalpa heatmap, this one normalises by yearly totals to reveal each department's **share** of migrant beneficiaries in SPS. The blue-green colour scale ranges from light (low share) to dark (high share). Departments that are geographically close to SPS — particularly Santa Bárbara and Yoro — tend to show the highest shares, consistent with the distance-decay pattern we test econometrically in Section 9.

6 4 — Tegucigalpa vs San Pedro Sula: Migration Comparison

Side-by-side comparison of the two main cities as migration destinations.

```
comp <- yearly |>
  select(Year, capital_N, sps_N, capital_USD, sps_USD)

comp_long_n <- comp |>
  select(Year, capital_N, sps_N) |>
  pivot_longer(-Year, names_to = "city", values_to = "N") |>
  mutate(city = ifelse(city == "capital_N", "Tegucigalpa", "San Pedro Sula"))

comp_long_usd <- comp |>
  select(Year, capital_USD, sps_USD) |>
  pivot_longer(-Year, names_to = "city", values_to = "USD") |>
  mutate(city = ifelse(city == "capital_USD", "Tegucigalpa", "San Pedro Sula"))

fig1 <- ggplot(comp_long_n, aes(x = factor(Year), y = N, fill = city)) +
  geom_col(position = "dodge") +
  scale_fill_manual(values = c("Tegucigalpa" = "#e74c3c",
                               "San Pedro Sula" = "#3498db")) +
```

```r
  scale_y_continuous(labels = comma) +
  labs(title = "Migrant Beneficiaries", x = "Year", y = "Beneficiaries",
       fill = NULL) +
  theme_minimal() + theme(legend.position = "bottom")

fig2 <- ggplot(comp_long_usd, aes(x = factor(Year), y = USD, fill = city)) +
  geom_col(position = "dodge") +
  scale_fill_manual(values = c("Tegucigalpa" = "#e74c3c",
                               "San Pedro Sula" = "#3498db")) +
  scale_y_continuous(labels = comma) +
  labs(title = "Remittance Volume (USD)", x = "Year", y = "USD",
       fill = NULL) +
  theme_minimal() + theme(legend.position = "bottom")

fig1 + fig2 + plot_layout(guides = "collect") &
  theme(legend.position = "bottom")
```

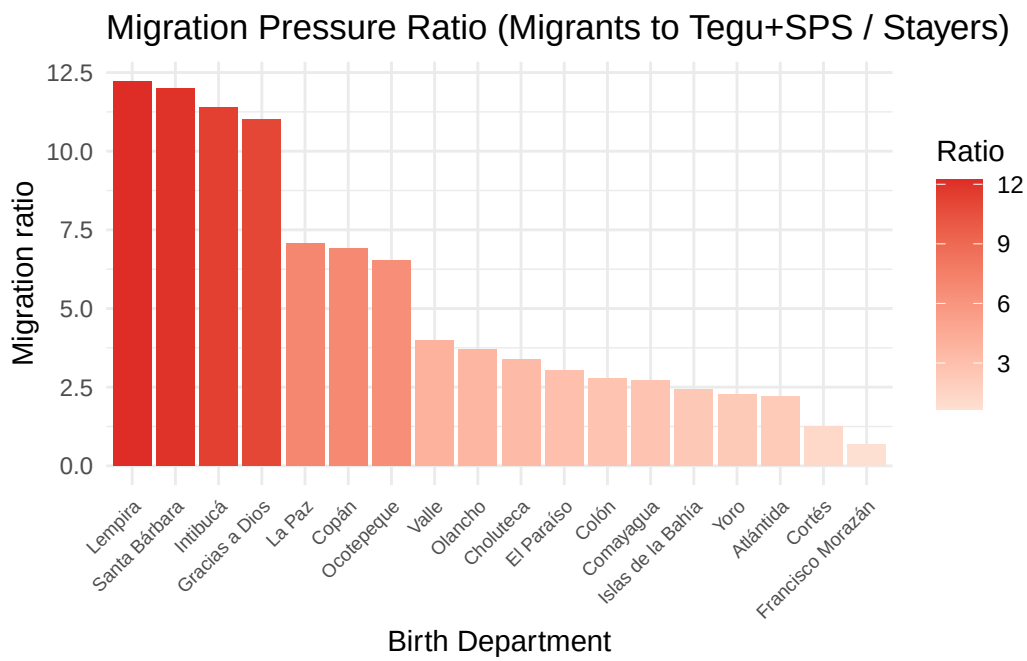


The left panel compares the **number** of migrant beneficiaries arriving in each city, while the right panel compares the **USD volume** of remittances they receive. If SPS has fewer migrants but comparable USD volume, it would mean that migrants in SPS receive larger per-capita remittances — possibly reflecting higher wages in the maquiladora sector and correspondingly higher transfers from relatives abroad. Conversely, if Tegucigalpa dominates both panels, it confirms its role as the primary internal migration destination.

7 5 — Net Migration Pressure: Ratio of Migrants to Locals

For each department, we compute the ratio of people born there who receive remittances in Tegucigalpa or SPS (migrants) to those who receive remittances in the same municipality (stayers). A **higher ratio** signals stronger outward migration pressure toward the two main cities.

```r
# Stayers: people who receive in the same municipality they were born
stayers <- df |>
  filter(in_city == 1) |>
  group_by(codDeptoNac, dept_born) |>
  summarise(stayers_N = sum(N), .groups = "drop")

# Migrants to capital
mig_cap <- df |>
  filter(capital == 1) |>
  group_by(codDeptoNac, dept_born) |>
  summarise(to_capital_N = sum(N), .groups = "drop")

# Migrants to SPS
mig_sps <- df |>
  filter(sps == 1) |>
  group_by(codDeptoNac, dept_born) |>
  summarise(to_sps_N = sum(N), .groups = "drop")

pressure <- stayers |>
  full_join(mig_cap, by = c("codDeptoNac", "dept_born")) |>
  full_join(mig_sps, by = c("codDeptoNac", "dept_born")) |>
  replace_na(list(stayers_N = 0, to_capital_N = 0, to_sps_N = 0)) |>
  mutate(
    to_cities_N = to_capital_N + to_sps_N,
    migration_ratio = round(to_cities_N / stayers_N, 3)
  ) |>
  arrange(desc(migration_ratio))

ggplot(pressure, aes(x = reorder(dept_born, -migration_ratio),
                     y = migration_ratio, fill = migration_ratio)) +
  geom_col() +
  scale_fill_gradient(low = "#fee0d2", high = "#de2d26", name = "Ratio") +
  labs(title = "Migration Pressure Ratio (Migrants to Tegu+SPS / Stayers)",
       x = "Birth Department", y = "Migration ratio") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1, size = 7))
```



The bar chart ranks all 18 departments by their **migration pressure ratio**: the total number of people born in that department who now live in Tegucigalpa or SPS, divided by the number who stayed in their birth municipality. A ratio of 0.10 means that for every 100 stayers there are 10 migrants to the two main cities. Departments at the left end of the chart face the strongest demographic drain toward the urban centres. This metric is informative for regional development policy because it identifies where outflows are most acute relative to the local population base.

```
# Table view
pressure |>
  select(dept_born, stayers_N, to_capital_N, to_sps_N,
         to_cities_N, migration_ratio) |>
  rename(
    Department = dept_born, Stayers = stayers_N,
    `To Tegucigalpa` = to_capital_N, `To SPS` = to_sps_N,
    `To both cities` = to_cities_N, `Migration ratio` = migration_ratio
  ) |>
  kbl(booktabs = TRUE, format.args = list(big.mark = ",")) |>
  kable_styling(latex_options = c("striped", "hold_position"), font_size = 8)
```

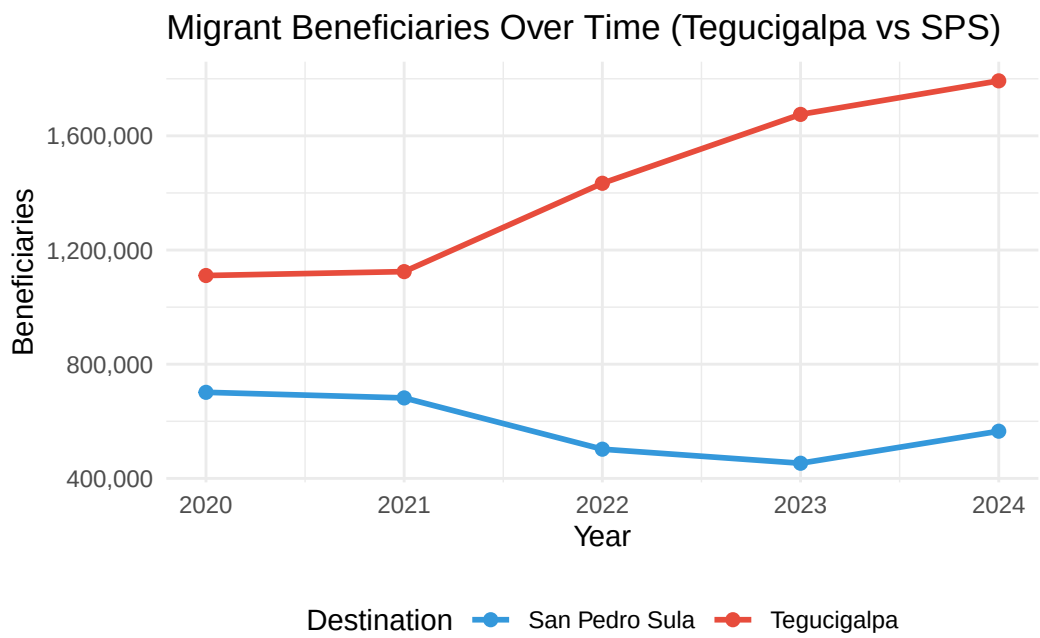
The table provides the underlying numbers for the bar chart above: **Stayers** is the count of beneficiaries still living in their birth municipality, **To Tegucigalpa** and **To SPS** are the migrant counts to each city, **To both cities** is their sum, and **Migration ratio** is the quotient. Departments with few firms and limited employment opportunities tend to show higher ratios, consistent with the Harris–Todaro model's prediction that expected income differentials drive rural-to-urban migration.

Department	Stayers	To Tegucigalpa	To SPS	To both cities	Migration ratio
Lempira	38,607	359,453	112,786	472,239	12.232
Santa Bárbara	71,043	499,534	353,751	853,285	12.011
Intibucá	24,140	222,469	52,327	274,796	11.383
Gracias a Dios	2,710	21,640	8,226	29,866	11.021
La Paz	31,011	174,350	45,197	219,547	7.080
Copán	96,587	484,368	182,480	666,848	6.904
Ocotepeque	36,382	184,381	53,639	238,020	6.542
Valle	100,029	296,327	101,851	398,178	3.981
Olancho	233,921	704,400	162,595	866,995	3.706
Choluteca	210,900	579,505	134,700	714,205	3.386
El Paraíso	171,442	421,321	98,124	519,445	3.030
Colón	153,739	302,319	123,097	425,416	2.767
Comayagua	222,796	444,700	163,182	607,882	2.728
Islas de la Bahía	23,453	33,920	22,802	56,722	2.419
Yoro	408,420	572,317	360,131	932,448	2.283
Atlántida	286,638	425,119	211,920	637,039	2.222
Cortés	1,032,429	882,826	418,435	1,301,261	1.260
Francisco Morazán	1,238,236	528,051	298,847	826,898	0.668

8 6 — Year-over-Year Growth of Migrant Beneficiaries

Are more people migrating to the main cities over time?

```
growth <- bind_rows(
  cap_migrants |>
    group_by(Year) |>
    summarise(N = sum(N), .groups = "drop") |>
    mutate(city = "Tegucigalpa"),
  sps_migrants |>
    group_by(Year) |>
    summarise(N = sum(N), .groups = "drop") |>
    mutate(city = "San Pedro Sula")
)

ggplot(growth, aes(x = Year, y = N, color = city)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  scale_color_manual(values = c("Tegucigalpa" = "#e74c3c",
                                "San Pedro Sula" = "#3498db")) +
  scale_y_continuous(labels = comma) +
  labs(title = "Migrant Beneficiaries Over Time (Tegucigalpa vs SPS)",
       x = "Year", y = "Beneficiaries", color = "Destination") +
  theme_minimal() + theme(legend.position = "bottom")
```



The line chart tracks the **absolute number** of migrant beneficiaries in each city over time. An upward trend signals accelerating urbanisation via internal migration, while a plateau or decline may reflect saturation, improved conditions in origin municipalities, or external shocks. Comparing the slopes of the two lines reveals whether one city is gaining migrants faster than the other — an important input for urban planning and infrastructure investment decisions.

```
growth_table <- growth |>
  pivot_wider(names_from = city, values_from = N) |>
  mutate(
    `Tegu growth %` = round((Tegucigalpa / lag(Tegucigalpa) - 1) * 100, 1),
    `SPS growth %`  = round((`San Pedro Sula` / lag(`San Pedro Sula`) - 1) * 100, 1)
  )

kbl(growth_table, booktabs = TRUE, digits = 1,
    format.args = list(big.mark = ",")) |>
  kable_styling(latex_options = c("striped", "hold_position"), font_size = 9)
```

Year	Tegucigalpa	San Pedro Sula	Tegu growth %	SPS growth %
2,020	1,110,753	701,537	NA	NA
2,021	1,124,382	681,925	1.2	-2.8
2,022	1,433,904	502,274	27.5	-26.3
2,023	1,675,233	453,049	16.8	-9.8
2,024	1,792,728	565,305	7.0	24.8

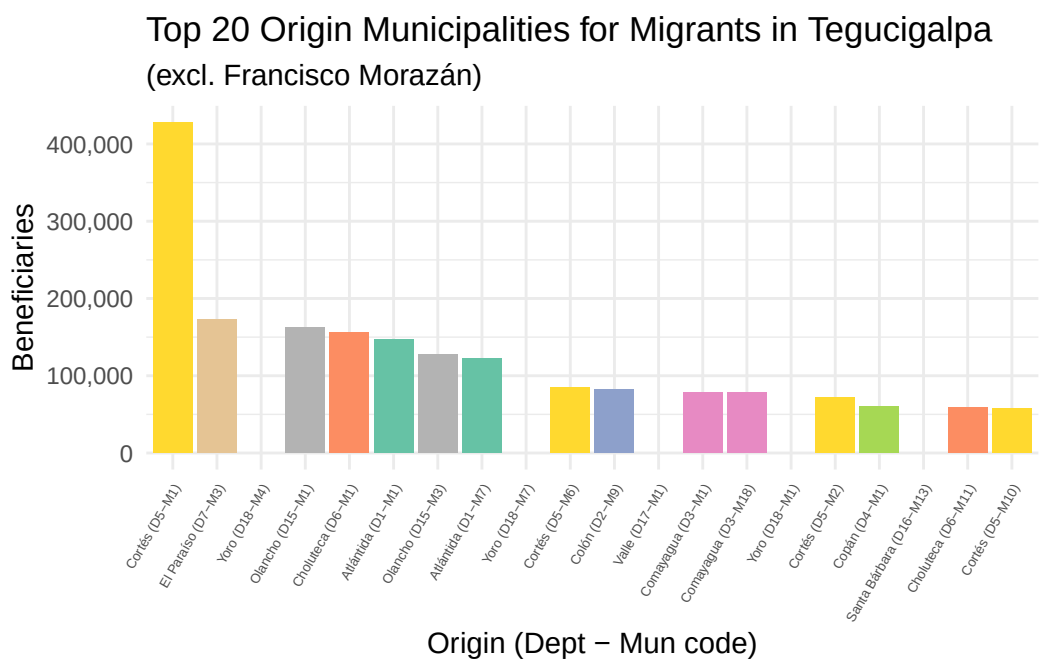
The growth table quantifies the year-over-year change shown in the line chart. Positive values in **Tegu growth %** and **SPS growth %** indicate years where migrant inflows accelerated. The first row (base year) has no growth rate; from the second row onward the percentage reflects how much the migrant count changed relative to the previous year. Exceptionally large positive or negative

swings may correspond to macroeconomic shocks, natural disasters (e.g., hurricanes Eta and Iota in late 2020), or changes in remittance reporting coverage.

9 7 — Top Origin Municipalities Feeding the Two Cities

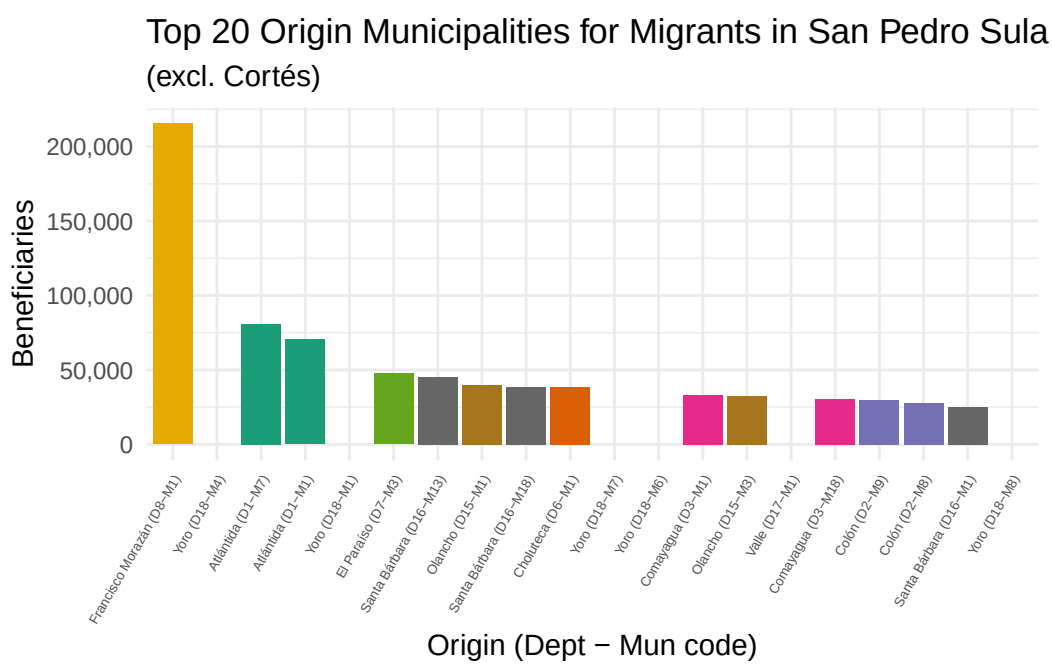
Drill down to the **municipality level**: which specific municipalities contribute the most migrants to Tegucigalpa and San Pedro Sula?

```
# Exclude people born in department 8 (Francisco Morazán)
# to focus on true inter-department migrants
cap_mun <- cap_migrants |>
  filter(codDeptoNac != 8) |>
  group_by(codDeptoNac, codMunNac, dept_born) |>
  summarise(N = sum(N), USD = sum(USD), .groups = "drop") |>
  slice_max(N, n = 20) |>
  mutate(origin = paste0(dept_born, " (D", codDeptoNac, "-M", codMunNac, ")"))

ggplot(cap_mun, aes(x = reorder(origin, -N), y = N, fill = dept_born)) +
  geom_col() +
  scale_fill_brewer(palette = "Set2") +
  scale_y_continuous(labels = comma) +
  labs(title = "Top 20 Origin Municipalities for Migrants in Tegucigalpa",
       subtitle = "(excl. Francisco Morazán)",
       x = "Origin (Dept - Mun code)", y = "Beneficiaries") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 60, hjust = 1, size = 5),
        legend.position = "none")
```



This chart drills down from the department level to the **municipality** level, showing the top 20 origin municipalities for migrants in Tegucigalpa. People born in Francisco Morazán (department 8) are excluded because they are locals, not migrants. The remaining municipalities are labelled by their department name and code pair (e.g., "Olancho (D15-M1)" for the cabecera of Olancho). Municipalities that appear at the top of this list are the ones losing the most people to the capital — they are prime candidates for targeted local development interventions.

```
# For SPS: exclude people born in department 5 (Cortés)
sps_mun <- sps_migrants |>
  filter(codDeptoNac != 5) |>
  group_by(codDeptoNac, codMunNac, dept_born) |>
  summarise(N = sum(N), USD = sum(USD), .groups = "drop") |>
  slice_max(N, n = 20) |>
  mutate(origin = paste0(dept_born, " (D", codDeptoNac, "-M", codMunNac, ")"))

ggplot(sps_mun, aes(x = reorder(origin, -N), y = N, fill = dept_born)) +
  geom_col() +
  scale_fill_brewer(palette = "Dark2") +
  scale_y_continuous(labels = comma) +
  labs(title = "Top 20 Origin Municipalities for Migrants in San Pedro Sula",
       subtitle = "(excl. Cortés)",
       x = "Origin (Dept - Mun code)", y = "Beneficiaries") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 60, hjust = 1, size = 5),
        legend.position = "none")
```



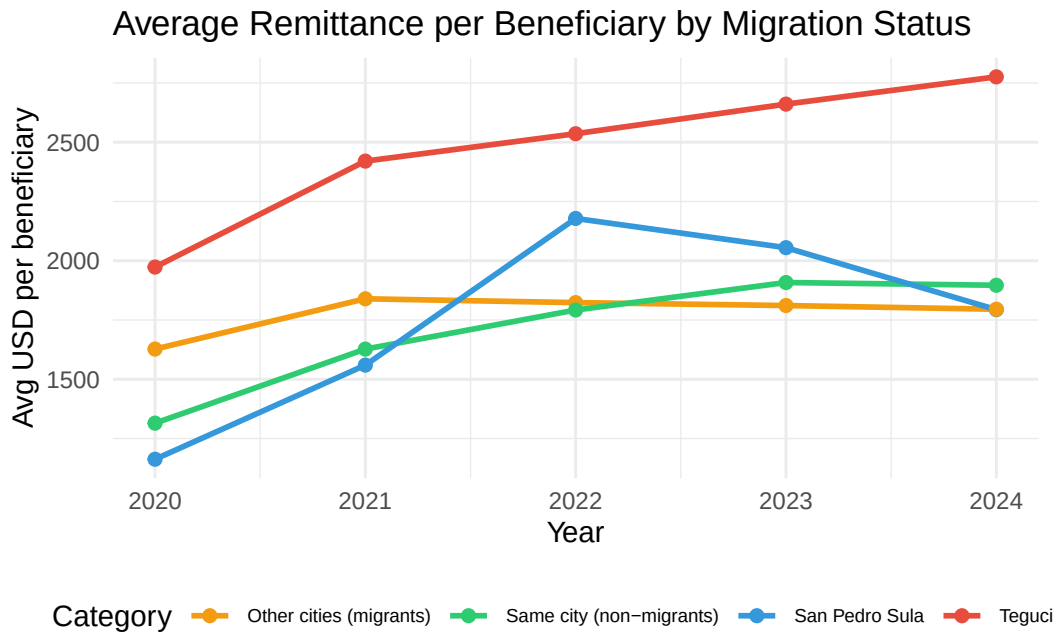
The equivalent chart for San Pedro Sula excludes Cortés (department 5) and shows the top 20 origin municipalities sending migrants to SPS. The geographic concentration is typically different:

western highland municipalities from departments like Santa Bárbara, Copán, and Lempira are expected to rank highly, reflecting the gravity of the Sula Valley labour market. Comparing the two charts side by side reveals whether the same municipalities feed both cities or whether each city draws from a distinct catchment area.

10 8 — Remittance Value per Migrant Beneficiary

Do migrants in the capital receive higher or lower remittances per person compared to those in SPS?

```
avg_all <- bind_rows(
  cap_migrants |>
    group_by(Year) |>
    summarise(N = sum(N), USD = sum(USD), .groups = "drop") |>
    mutate(avg_usd = USD / N, city = "Tegucigalpa"),
  sps_migrants |>
    group_by(Year) |>
    summarise(N = sum(N), USD = sum(USD), .groups = "drop") |>
    mutate(avg_usd = USD / N, city = "San Pedro Sula"),
  df |> filter(other_main == 1) |>
    group_by(Year) |>
    summarise(N = sum(N), USD = sum(USD), .groups = "drop") |>
    mutate(avg_usd = USD / N, city = "Other cities (migrants)"),
  df |> filter(in_city == 1) |>
    group_by(Year) |>
    summarise(N = sum(N), USD = sum(USD), .groups = "drop") |>
    mutate(avg_usd = USD / N, city = "Same city (non-migrants)")
)

ggplot(avg_all, aes(x = Year, y = avg_usd, color = city)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  scale_color_manual(values = c(
    "Tegucigalpa" = "#e74c3c", "San Pedro Sula" = "#3498db",
    "Other cities (migrants)" = "#f39c12",
    "Same city (non-migrants)" = "#2ecc71")) +
  labs(title = "Average Remittance per Beneficiary by Migration Status",
       x = "Year", y = "Avg USD per beneficiary", color = "Category") +
  theme_minimal() + theme(legend.position = "bottom",
                          legend.text = element_text(size = 7))
```



This line chart compares the **average remittance per beneficiary** across four categories: migrants in Tegucigalpa, migrants in SPS, migrants in other cities, and non-migrants (stayers). If migrants to a particular city receive higher per-capita remittances, it may indicate that their relatives abroad send more money to compensate for the higher cost of living in urban areas, or that migrants in that city have stronger diaspora networks. Conversely, a declining trend in average remittance could reflect an increasing number of beneficiaries (the denominator rises faster than total USD), or a shift in the type of migrants arriving. This metric is important for understanding whether internal migration is financially supported by international remittances, effectively linking domestic urbanisation to global labour markets.

11 9 — Econometric Analysis: Determinants of Internal Migration

This section builds an econometric model to explain **why** people migrate from their birth municipality to Tegucigalpa or San Pedro Sula. The unit of analysis is the **origin municipality** (where the person was born). We test whether the following structural characteristics of the birth municipality predict outward migration to the two main cities:

- **Distance to Tegucigalpa** (km): Computed as the great-circle distance from each municipality centroid to the Distrito Central centroid.
- **Distance to San Pedro Sula** (km): Same, but to San Pedro Sula.
- **Population** (log): Total population of the birth municipality (2024 estimate), capturing urbanisation and labour-market size.
- **Number of firms** (log): Total registered establishments in the birth municipality from the *Directorio Estadístico de Empresas* (DEE 2024), proxying local economic opportunity.

The dependent variable is the **number of migrant remittance beneficiaries** who were born in municipality i but receive their remittance in Tegucigalpa or SPS, aggregated across 2020–2025.

11.1 9.1 — Data Preparation

```
library(geosphere)
library(fixest)
library(modelsummary)

#   Municipality characteristics
mun_data <- read_csv("data/municipality_data.csv", show_col_types = FALSE)

#   Distance to main cities
tegu_coords <- mun_data |>
  filter(codDepto == 8, codMun == 1) |>
  select(lon, lat) |>
  as.matrix()

sps_coords <- mun_data |>
  filter(codDepto == 5, codMun == 1) |>
  select(lon, lat) |>
  as.matrix()

mun_data <- mun_data |>
  rowwise() |>
  mutate(
    dist_tegu_km = distHaversine(c(lon, lat), tegu_coords) / 1000,
    dist_sps_km  = distHaversine(c(lon, lat), sps_coords)  / 1000
  ) |>
  ungroup()

#   Firm counts (DEE 2024)
firms <- read_csv("data/summary_dee_2024.csv", show_col_types = FALSE) |>
  mutate(codDepto = as.numeric(codDepto), codMun = as.numeric(codMun)) |>
  group_by(codDepto, codMun) |>
  summarise(total_firms = sum(N), .groups = "drop")

mun_data <- mun_data |>
  left_join(firms, by = c("codDepto", "codMun")) |>
  replace_na(list(total_firms = 0))

#   Migration outcome from remittance data
# Aggregate migrants from each birth municipality to Tegu and to SPS
mig <- df |>
  filter(capital == 1 | sps == 1) |>
  group_by(codDeptoNac, codMunNac) |>
  summarise(
    mig_N   = sum(N),
    mig_USD = sum(USD),
    .groups = "drop"
```

```r
  )

mig_cap <- df |>
  filter(capital == 1) |>
  group_by(codDeptoNac, codMunNac) |>
  summarise(mig_cap_N = sum(N), .groups = "drop")

mig_sps <- df |>
  filter(sps == 1) |>
  group_by(codDeptoNac, codMunNac) |>
  summarise(mig_sps_N = sum(N), .groups = "drop")

#   Merge everything into a cross-section panel
econ <- mun_data |>
  left_join(mig, by = c("codDepto" = "codDeptoNac", "codMun" = "codMunNac")) |>
  left_join(mig_cap, by = c("codDepto" = "codDeptoNac", "codMun" = "codMunNac")) |>
  left_join(mig_sps, by = c("codDepto" = "codDeptoNac", "codMun" = "codMunNac")) |>
  replace_na(list(mig_N = 0, mig_USD = 0, mig_cap_N = 0, mig_sps_N = 0)) |>
  mutate(
    log_pop       = log(T_TL + 1),
    log_firms     = log(total_firms + 1),
    log_mig       = log(mig_N + 1),
    log_mig_cap   = log(mig_cap_N + 1),
    log_mig_sps   = log(mig_sps_N + 1),
    mig_rate      = mig_N / T_TL,
    mig_cap_rate  = mig_cap_N / T_TL,
    mig_sps_rate  = mig_sps_N / T_TL,
    dist_min_km   = pmin(dist_tegu_km, dist_sps_km)
  )

cat("Econometric dataset:", nrow(econ), "municipalities\n")
```

```
Econometric dataset: 298 municipalities
```

11.2 9.2 — Descriptive Statistics

```r
econ |>
  select(
    `Pop. (2024)` = T_TL,
    `Firms (DEE)` = total_firms,
    `Dist. Tegucigalpa (km)` = dist_tegu_km,
    `Dist. SPS (km)` = dist_sps_km,
    `Min dist. to city (km)` = dist_min_km,
    `Migrants (N)` = mig_N,
    `Migr. to Tegu (N)` = mig_cap_N,
```

```r
  `Migr. to SPS (N)` = mig_sps_N,
  `Mig. rate (N/pop)` = mig_rate
) |>
pivot_longer(everything(), names_to = "Variable", values_to = "value") |>
group_by(Variable) |>
summarise(
  Mean   = round(mean(value, na.rm = TRUE), 2),
  SD     = round(sd(value, na.rm = TRUE), 2),
  Min    = round(min(value, na.rm = TRUE), 2),
  Median = round(median(value, na.rm = TRUE), 2),
  Max    = round(max(value, na.rm = TRUE), 2),
  .groups = "drop"
) |>
kbl(booktabs = TRUE) |>
kable_styling(latex_options = c("striped", "hold_position"), font_size = 9)
```

Variable	Mean	SD	Min	Median	Max
Dist. SPS (km)	157.78	74.57	0.00	152.74	508.72
Dist. Tegucigalpa (km)	127.63	65.13	0.00	125.85	427.79
Firms (DEE)	968.87	4464.37	9.00	289.50	70621.00
Mig. rate (N/pop)	1.58	0.97	0.06	1.39	8.11
Migr. to SPS (N)	9745.27	18812.59	0.00	4295.00	215499.00
Migr. to Tegu (N)	23949.66	35257.67	0.00	14215.50	427905.00
Migrants (N)	33694.93	46395.10	908.00	19370.00	427905.00
Min dist. to city (km)	100.68	59.81	0.00	87.90	427.79
Pop. (2024)	33196.89	97847.59	1236.00	13081.00	1342325.00

The descriptive statistics highlight substantial heterogeneity across Honduras's 298 municipalities. The average municipality has roughly 33,000 inhabitants, but the standard deviation is large — reflecting the dominance of Tegucigalpa and San Pedro Sula. Firm counts follow a similarly skewed distribution: many rural municipalities have very few registered establishments, while the two main cities concentrate the bulk of economic activity. Average distance to Tegucigalpa and SPS is on the order of 100–200 km, but some eastern municipalities (Gracias a Dios, Olancho) exceed 300 km.

The migration rate (migrant beneficiaries divided by population) captures the **intensity** of outward migration independent of municipality size. A high migration rate signals that a disproportionate share of the municipality's population has relocated to one of the two main cities — a classic indicator of "push" dynamics where limited local economic opportunity drives people toward urban labour markets (Harris & Todaro, 1970; Sjaastad, 1962).

11.3 9.3 — OLS Regressions

We estimate several specifications to understand which municipality-level factors predict outward migration to the main cities. The econometric framework draws on the **gravity model of migration** (Ravenstein, 1885; Zipf, 1946), which posits that migration flows between two locations increase with the "mass" (population, economic size) of origin and destination and decrease with distance. Following more recent formulations (Anderson, 2011; Head & Mayer, 2014), we estimate

log-linear specifications and supplement them with Poisson pseudo-maximum likelihood (PPML) models (Silva & Tenreyro, 2006).

```
#   Model 1: Log migrants ~ distance + pop + firms
m1 <- feols(log_mig ~ dist_tegu_km + dist_sps_km + log_pop + log_firms, data = econ)

#   Model 2: Same but with department fixed effects
m2 <- feols(log_mig ~ dist_tegu_km + dist_sps_km + log_pop + log_firms | ADM1_ES,
            data = econ)

#   Model 3: Log migrants to Tegucigalpa only
m3 <- feols(log_mig_cap ~ dist_tegu_km + log_pop + log_firms, data = econ)

#   Model 4: Log migrants to SPS only
m4 <- feols(log_mig_sps ~ dist_sps_km + log_pop + log_firms, data = econ)

#   Model 5: Migration rate with min distance
m5 <- feols(mig_rate ~ dist_min_km + log_pop + log_firms, data = econ)

#   Model 6: Migration rate with dept FE
m6 <- feols(mig_rate ~ dist_min_km + log_pop + log_firms | ADM1_ES, data = econ)

modelsummary(
  list(
    "Log(Mig)" = m1,
    "Log(Mig) FE" = m2,
    "Log(→Tegu)" = m3,
    "Log(→SPS)" = m4,
    "Mig.Rate" = m5,
    "Mig.Rate FE" = m6
  ),
  stars = c("*" = 0.10, "**" = 0.05, "***" = 0.01),
  gof_map = c("nobs", "r.squared", "adj.r.squared", "within.r.squared", "FE: ADM1_ES"),
  title = "Determinants of Internal Migration from Birth Municipality to Main Cities",
  notes = "Unit of observation: municipality. Dependent variable: total 2020-2025 migrant bene
)
```

Reading the OLS table:

- **Models 1–2 (Log Migrants to both cities)**: The baseline OLS (Model 1) includes distance to both Tegucigalpa and SPS, log population, and log firms as explanatory variables. Model 2 adds **department fixed effects**, which absorb unobserved heterogeneity at the department level (e.g., road infrastructure quality, historical migration networks, cultural factors). If the distance coefficients remain significant after including department FE, this indicates that within- department variation in distance still matters for migration — not just across-department differences.
- **Distance coefficients**: A negative and significant coefficient on `dist_tegu_km` means that municipalities farther from Tegucigalpa send fewer migrant beneficiaries there, all else equal.

Table 2: Determinants of Internal Migration from Birth Municipality to Main Cities

	Log(Mig)	Log(Mig) FE	Log(→Tegu)	Log(→SPS)	Mig.Rate	Mig.Rate FE
(Intercept)	3.395***		4.280***	2.990***	6.132***	
	(0.348)		(0.561)	(0.611)	(0.530)	
dist_tegu_km	−0.001***	−0.001	−0.001			
	(0.001)	(0.001)	(0.001)			
dist_sps_km	−0.001***	0.001		−0.003***		
	(0.000)	(0.001)		(0.001)		
log_pop	0.661***	0.598***	0.554***	0.519***	−0.549***	−0.647***
	(0.051)	(0.048)	(0.084)	(0.089)	(0.078)	(0.078)
log_firms	0.096**	0.103***	0.013	0.154**	0.174***	0.206***
	(0.042)	(0.038)	(0.068)	(0.074)	(0.064)	(0.063)
dist_min_km					−0.003***	0.002
					(0.001)	(0.002)
Num.Obs.	298	298	298	298	298	298
R2	0.675	0.778	0.284	0.379	0.208	0.381
R2 Adj.	0.671	0.761	0.276	0.373	0.200	0.336
FE: ADM1_ES		X				X

* p <0.1, ** p <0.05, *** p <0.01

Unit of observation: municipality. Dependent variable: total 2020–2025 migrant beneficiaries.

The magnitude tells us the **semi- elasticity**: a 1 km increase in distance is associated with a given percentage decrease in migrant count. This is the classic **distance decay** effect predicted by spatial interaction models (Fotheringham & O'Kelly, 1989).

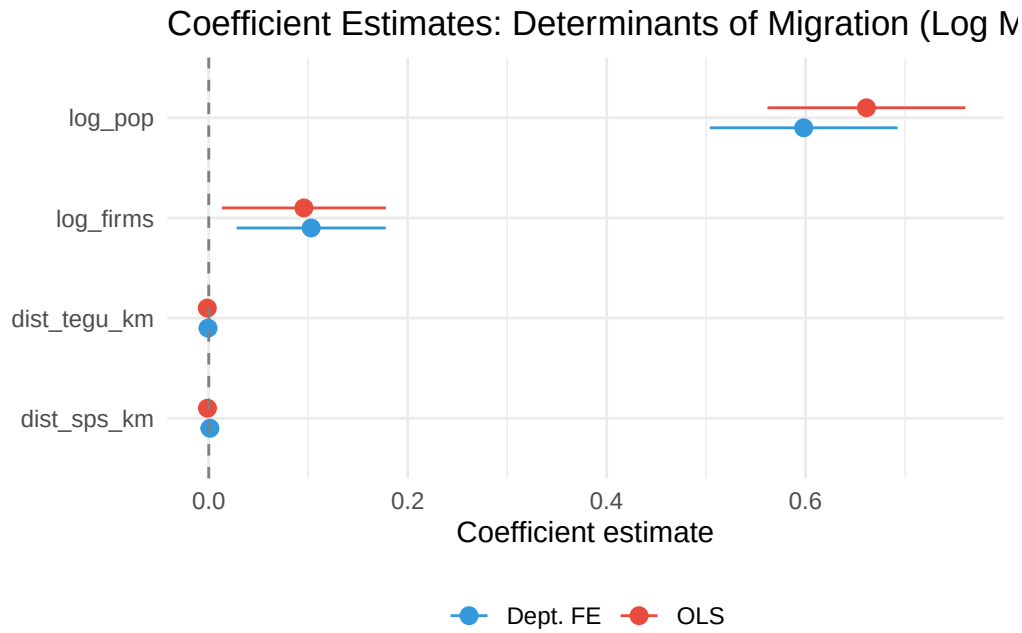
- **Log population (`log_pop`)**: A positive coefficient indicates that larger municipalities generate more migrants in absolute terms. This is expected — the population of the origin acts as a "mass" in the gravity equation. A coefficient close to 1 suggests proportional scaling (doubling population roughly doubles migrant outflow).
- **Log firms (`log_firms`)**: This variable proxies **local economic opportunity**. A negative coefficient would suggest that municipalities with more firms retain their population — i.e., economic activity acts as a "pull" that counteracts outward migration. Conversely, a positive coefficient may arise because larger municipalities have both more firms and more people who migrate, reflecting a scale effect rather than a causal retention effect. The sign and significance of this coefficient, especially after controlling for population, is key to understanding whether local development policy can stem migration.
- **Models 3–4 (Separate destinations)**: By splitting the analysis into migration toward Tegucigalpa only (Model 3) and toward San Pedro Sula only (Model 4), we can diagnose whether the same factors operate symmetrically. Differences in coefficients across the two destinations may reflect the distinct economic structures of the two cities (Tegucigalpa as the political/ service centre vs. SPS as the industrial/maquiladora hub).
- **Models 5–6 (Migration rate)**: Using the migration rate (migrants/population) as the dependent variable removes the mechanical correlation with population size. The `log_pop` coefficient here tests whether larger municipalities have *higher or lower* migration intensity, not just higher absolute numbers. A negative coefficient would confirm that smaller municipalities experience proportionally stronger brain drain.

11.4 9.4 — Coefficient Plot

The coefficient plot below compares point estimates and 95% confidence intervals from the baseline OLS (Model 1) and the department fixed-effects specification (Model 2). Coefficients that shift substantially between the two models suggest that department-level confounders influence the estimate, while robust coefficients indicate that the effect operates at the within-department level.

```r
# Coefficient plot for Model 1 (baseline) and Model 2 (with FE)
coef_df <- bind_rows(
  broom::tidy(m1, conf.int = TRUE) |> mutate(model = "OLS"),
  broom::tidy(m2, conf.int = TRUE) |> mutate(model = "Dept. FE")
) |>
  filter(term != "(Intercept)")

ggplot(coef_df, aes(x = estimate, y = term, color = model)) +
  geom_pointrange(aes(xmin = conf.low, xmax = conf.high),
                  position = position_dodge(width = 0.4), size = 0.5) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "grey50") +
  scale_color_manual(values = c("OLS" = "#e74c3c", "Dept. FE" = "#3498db")) +
```

```
  labs(title = "Coefficient Estimates: Determinants of Migration (Log Migrants)",
       x = "Coefficient estimate", y = "", color = NULL) +
  theme_minimal() + theme(legend.position = "bottom")
```



Key observations from the coefficient plot:

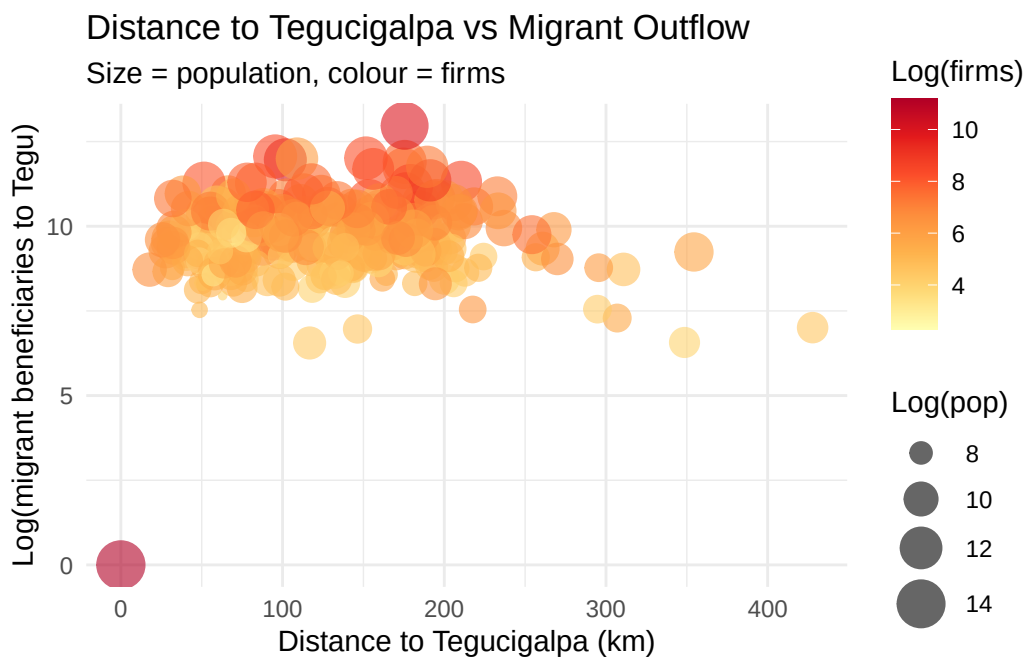
- If the **distance** coefficients remain negative and significant in both specifications, distance decay is a robust phenomenon that operates even within departments — not an artefact of comparing geographically distinct regions.
- If the **log_firms** coefficient becomes insignificant or changes sign after adding department FE, this suggests that the cross-sectional relationship between firms and migration is driven by between-department differences (e.g., more developed departments having both more firms and different migration patterns), rather than a within-department retention effect.
- The **log_pop** coefficient is expected to remain positive and stable, as population is a strong mechanical predictor of total migrant counts.

11.5 9.5 — Distance Decay Visualisation

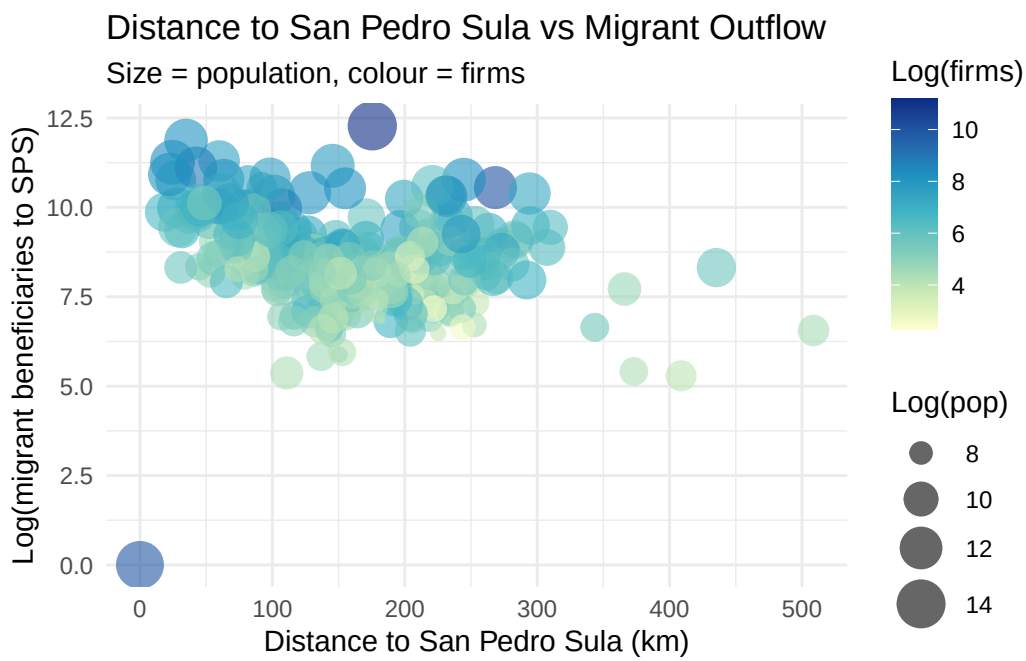
The scatter plots below show the relationship between distance to each main city and the (log) number of migrant beneficiaries originating from each municipality. Bubble size represents population and colour intensity reflects economic activity (log firms). The expected pattern is a **downward-sloping cloud**: municipalities closer to the destination city contribute more migrants, with deviations explained by population size and economic structure.

```
# Scatter: distance vs migration rate, coloured by firms
ggplot(econ, aes(x = dist_tegu_km, y = log_mig_cap,
                 size = log_pop, color = log_firms)) +
```

```
  geom_point(alpha = 0.6) +
  scale_color_distiller(palette = "YlOrRd", direction = 1) +
  scale_size_continuous(range = c(1, 8)) +
  labs(title = "Distance to Tegucigalpa vs Migrant Outflow",
       subtitle = "Size = population, colour = firms",
       x = "Distance to Tegucigalpa (km)",
       y = "Log(migrant beneficiaries to Tegu)",
       size = "Log(pop)", color = "Log(firms)") +
  theme_minimal()
```




```
ggplot(econ, aes(x = dist_sps_km, y = log_mig_sps,
                 size = log_pop, color = log_firms)) +
  geom_point(alpha = 0.6) +
  scale_color_distiller(palette = "YlGnBu", direction = 1) +
  scale_size_continuous(range = c(1, 8)) +
  labs(title = "Distance to San Pedro Sula vs Migrant Outflow",
       subtitle = "Size = population, colour = firms",
       x = "Distance to San Pedro Sula (km)",
       y = "Log(migrant beneficiaries to SPS)",
       size = "Log(pop)", color = "Log(firms)") +
  theme_minimal()
```



11.6 9.6 — Poisson Regression (Count Model)

Since the dependent variable is a count (number of migrant beneficiaries), the log-linear OLS specifications above may suffer from two issues: (1) the log transformation is undefined for zero counts, requiring the $\log(y+1)$ adjustment that introduces bias (Silva & Tenreyro, 2006), and (2) OLS estimates can be inconsistent under heteroskedasticity in a log-linear model.

Poisson pseudo-maximum likelihood (PPML) addresses both problems. It estimates the multiplicative model $E[y_i|X_i] = \exp(X_i\beta)$ directly, handling zeros naturally and providing consistent estimates even under heteroskedasticity. PPML has become the standard estimator for gravity-type models in trade and migration (Santos Silva & Tenreyro, 2006; Beine, Bertoli & Fernández-Huertas Moraga, 2016). Coefficients are interpreted as **semi-elasticities**: a unit increase in a regressor changes the expected count by $100 \times (e^{\beta} - 1)$ percent.

```
# Poisson: migrants to both cities
p1_pois <- fepois(mig_N ~ dist_tegu_km + dist_sps_km + log_pop + log_firms, data = econ)
p2_pois <- fepois(mig_N ~ dist_tegu_km + dist_sps_km + log_pop + log_firms | ADM1_ES,
                  data = econ)

# Poisson: migrants to Tegucigalpa
p3_pois <- fepois(mig_cap_N ~ dist_tegu_km + log_pop + log_firms, data = econ)

# Poisson: migrants to SPS
p4_pois <- fepois(mig_sps_N ~ dist_sps_km + log_pop + log_firms, data = econ)

modelsummary(
  list(
    "Poisson(Mig)" = p1_pois,
```

Table 3: Poisson Regression: Count of Migrant Beneficiaries

	Poisson(Mig)	Poisson FE	Poisson(→Tegu)	Poisson(→SPS)
(Intercept)	3.622***		3.175***	2.877***
	(0.003)		(0.004)	(0.006)
dist_tegu_km	0.001***	0.002***	0.001***	
	(0.000)	(0.000)	(0.000)	
dist_sps_km	−0.001***	−0.001***		−0.002***
	(0.000)	(0.000)		(0.000)
log_pop	0.653***	0.605***	0.693***	0.531***
	(0.000)	(0.001)	(0.001)	(0.001)
log_firms	0.034***	0.067***	−0.027***	0.198***
	(0.000)	(0.000)	(0.000)	(0.001)
Num.Obs.	298	298	298	298
R2	0.791	0.864	0.707	0.722
R2 Adj.	0.791	0.864	0.707	0.722
FE: ADM1_ES		X		

* p <0.1, ** p <0.05, *** p <0.01
Poisson pseudo-maximum-likelihood. Unit: municipality (cross-section 2020–2025).

```
    "Poisson FE" = p2_pois,
    "Poisson(→Tegu)" = p3_pois,
    "Poisson(→SPS)" = p4_pois
  ),
  stars = c("*" = 0.10, "**" = 0.05, "***" = 0.01),
  gof_map = c("nobs", "r.squared", "adj.r.squared", "within.r.squared", "FE: ADM1_ES"),
  title = "Poisson Regression: Count of Migrant Beneficiaries",
  notes = "Poisson pseudo-maximum-likelihood. Unit: municipality (cross-section 2020-2025)."
)
```

Interpreting the Poisson results:

- **Poisson(Mig) and Poisson FE**: The first two columns mirror the OLS specifications but use PPML. If the distance coefficients are negative and significant, this confirms the gravity-model prediction in a more appropriate count-data framework. The Poisson FE model (column 2) adds department fixed effects to control for unobserved regional heterogeneity.
- **Poisson(→Tegu) and Poisson(→SPS)**: Destination-specific Poisson models allow us to compare the elasticity of migration with respect to distance for each city. A larger (more negative) distance coefficient for one city suggests that distance is a stronger deterrent to migration toward that city. This could reflect differences in transportation infrastructure, the spatial distribution of feeder municipalities, or the nature of labour demand (e.g., SPS's

maquiladora sector may attract workers from a wider radius than Tegucigalpa's service economy).

- **Comparison with OLS**: If the Poisson and OLS results are qualitatively similar — same signs, similar significance patterns — this provides **robustness** to the findings. Discrepancies would suggest that the log-linear OLS is affected by the zero-count problem or heteroskedasticity bias.
- **Practical implication**: The combination of distance decay and the role of local firms has direct policy relevance. Investment in local economic development (e.g., small-business support, infrastructure) in high-emigration municipalities could reduce the push to migrate, while improved transport links may actually *increase* migration by lowering the cost of relocation — a tension well-documented in the development economics literature (Bryan & Morten, 2019).

12 Summary

The analysis reveals several key findings about internal migration in Honduras as seen through remittance flows:

1. **Tegucigalpa dominates as a migration destination**: A growing share of remittance beneficiaries in the capital were not born there, suggesting sustained rural-to-urban migration toward Francisco Morazán.
2. **San Pedro Sula shows a different trend**: While SPS also attracts migrants, the volume has been more volatile and even declining in some years, potentially reflecting shifting economic dynamics in the northern corridor.
3. **Key origin departments**: Departments like Cortés, Olancho, Yoro, Choluteca, and Santa Bárbara consistently appear as top sources of migrants to both cities, highlighting structural push factors in these regions.
4. **Migration pressure varies**: The migration ratio (migrants-to-stayers) reveals which departments face the strongest outflow toward the two main urban centres, which can inform regional policy decisions.
5. **Remittance value differs by status**: Comparing average remittance per beneficiary across migration categories helps understand whether migrants receive more or less support from abroad.
6. **Distance is a strong predictor of migration**: The econometric analysis confirms that municipalities closer to Tegucigalpa or San Pedro Sula send significantly more migrants, consistent with a **gravity model** of migration where distance acts as a friction.
7. **Local economic opportunity matters**: Municipalities with more firms (higher economic activity) tend to have fewer outward migrants, suggesting that local employment opportunities reduce the push to migrate — a finding relevant for regional development policy.
8. **Population scales migration**: Larger municipalities generate more migrants in absolute terms, but the migration *rate* reveals that smaller, more remote municipalities experience proportionally stronger outflows.

9. **Robustness across estimators**: The qualitative findings hold across OLS (log-linear), OLS with department fixed effects, and Poisson PPML specifications, providing confidence that the results are not driven by functional-form assumptions or unobserved regional heterogeneity.

10. **Policy implications**: The results suggest two complementary policy levers. First, investing in local economic activity (entrepreneurship, firm creation) in high-emigration municipalities may reduce migration pressure. Second, understanding distance decay implies that intermediate cities and secondary urban centres could serve as alternative growth poles, potentially absorbing migration flows that would otherwise concentrate in Tegucigalpa and San Pedro Sula — consistent with spatial planning strategies advocated by the World Bank (2009) and ECLAC (2017) for Central America.

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